

CarbonTally — Migration Manifest (RC1 Package)

This manifest is the implementation roadmap the Product Owner approves before migration packages are generated and executed **one at a time, in order, each gated by the verification suite (007)**. Platform: **Supabase / PostgreSQL 16**, schema `public`, UK primary launch with Ireland beta enabled structurally. The package comprises seven files (001–006 migrations, 007 read-only verification) plus release notes (008). **Standing precondition — Gate 1 (“action zero”)**: before any execution, the RC1 migration files must be inspected against the real Supabase migration history, because the schema dump showed no indexes, foreign keys, CHECKs or RLS policies and is known to be silent on them. Every RC1 file is verify-first and idempotent, so Gate 1 may convert some “added” counts below into “verified pre-existing” — that reconciliation is expected and is not a defect. One RC2 amendment is incorporated: the rename batch in migration 001 also covers `manual_extraction_items.defra_factor_used` → `emission_factor_used` (approved RC2-003, Architecture Freeze).

1. Execution Order and Dependencies

Order	File	Purpose	Depends on	Transactional?	Rollback posture
1	001_rc1_schema.sql	Renames (R1–R3 + RC2-003), 16 new columns, type widening, NOT NULL relaxation	Gate 1 inspection	Yes — single BEGIN/COMMIT	Inline commented reverse DDL per change; C7 reversal lossy if any value > int4 max
2	002_rc1_constraints.sql	CHECKs, UNIQUEs, FKs, NOT NULL tightenings	001’s renames and new columns must be in place (constraints reference final names: <code>emission_factors.country</code> , <code>emissions_logs.emission_factor_id</code> , <code>dpq.emission_factor_used</code> , <code>facilities.eircode</code>)	Yes — single BEGIN/COMMIT	Named constraints: DROP CONSTRAINT/INDEX per item; K7 backfill irreversible (pre-migration snapshot of the six tables is the only un-backfill path)
3	003_rc1_indexes.sql	18 indexes + 2 extensions	001/002 final names (indexes reference <code>emission_factor_id</code> , K4-aligned partial predicates)	No — non-transactional (CREATE INDEX CONCURRENTLY cannot run inside a transaction); run outside the transaction batch, statement-by-statement, never wrapped by a transaction-forcing	DROP INDEX CONCURRENTLY per index; a failed build leaves an INVALID index — drop and re-run; file is idempotent and re-runnable

Order	File	Purpose	Depends on	Transactional?	Rollback posture
				runner	
4	004_rc1_rls.sql	RLS enablement + ~160 policies + 2 helper functions	001 (final table/column names, <code>organizations.is_active</code> from C2)	Yes	Additive; full-file rollback template drops only policies it created; never drop pre-existing policies; prefer fixing forward
5	005_rc1_functions.sql	<code>set_updated_at()</code> , <code>anonymise_user()</code>	003's pgcrypto (sha256 for the erasure email hash); 001 final names	Yes	DROP FUNCTION per item; erasure itself is irreversible by design — Gate 5 staging rehearsal is the mitigation
6	006_rc1_triggers.sql	<code>trg_set_updated_at_*</code> on ≤76 mutable tables	005's <code>set_updated_at()</code> ; 001 final names	Yes	Commented DO block drops every owned trigger by name pattern
7	007_rc1_verification.sql	Read-only verification suite (10 check areas)	001–006 applied	n/a (SELECTs and NOTICE-only DO blocks)	Nothing to roll back; run last after each package and re-runnable at any time

Batching note: files 001 and 002 form one logical schema-and-constraint batch but are executed as **two separately reviewed steps** (002 cannot precede 001; 001 must verify clean before 002 starts). File 003 deliberately sits outside the transaction batch.

2. Migration 001 — Schema

Tables affected (9 modified, 1 renamed, 0 added): `defra_conversion_factors` → `emission_factors` (rename); modified: `facilities`, `organizations`, `consultant_billing`, `emission_factors`, `emissions_logs`, `customer_documents`, `file_attachments`, `suppliers`, `organization_metadata`, plus `manual_extraction_items` (RC2-003 rename target).

Columns added — 16, by table:

Table	Columns added
<code>facilities</code>	<code>eircode</code> (C1), <code>meter_mpan_mprn</code> (C9)
<code>organizations</code>	<code>is_active</code> (C2; NOT NULL DEFAULT true, backfilled true), <code>archived_at</code> (C2)
<code>consultant_billing</code>	<code>currency</code> (C3; DEFAULT 'GBP', backfilled)
<code>emission_factors</code>	<code>unit</code> , <code>scope</code> , <code>factor_source</code> , <code>factor_set</code> , <code>country</code> (C4 provenance set; backfilled DEFRA-DESNZ / GB / DEFRA-<year>)
<code>emissions_logs</code>	<code>unit</code> , <code>scope</code> (C5; derived backfill via factor join)

Table	Columns added
customer_documents	file_checksum (C6; SHA-256 hex, deliberately not unique)
suppliers	sort_code (C8)
organization_metadata	total_floor_area_sqm, occupied_floor_area_sqm (C10)

Columns renamed (4 renames + 1 retirement):

Old	New	Note
defra_conversion_factors (table)	emission_factors	R1 — jurisdiction-neutral v1.1 enabler
emissions_logs.defra_factor_id	emission_factor_id	R2
document_processing_queue.defra_factor_used	emission_factor_used	R2
manual_extraction_items.defra_factor_used	emission_factor_used	RC2 amendment (RC2-003) — added to this batch
organizations.default_defra_version	default_factor_year	R3
emission_factors.region	region_deprecated	C4 retirement — non-destructive; values folded into country first; drop only after consumer audit

Type changes (1): `file_attachments.file_size` `int4` → `int8` (C7 — aligns with the `int8` peer columns `document_processing_queue.file_size_bytes` / `processing_queue.file_size_bytes`; removes the 2 GB ceiling).

NOT NULL relaxation (1): `facilities.postcode` NOT NULL dropped (C1 — Ireland has no postcodes; the pairwise presence CHECK lands in 002).

3. Migration 002 — Constraints

All CHECKs and FKs on populated tables are added **NOT VALID then VALIDATE CONSTRAINT**, preceded by value-mapping UPDATES, so existing rows are never scanned under a write-blocking lock. No format/regex CHECKs anywhere (K9 rejected — API layer owns format validation). Staging data audit (NULL counts, duplicate sweeps, value mapping) runs before this file.

Group	Count	Target tables / names
Presence CHECK (C1 companion)	1	<code>facilities.postcode_or_eircode_check</code> — postcode OR eircode present
Country IN ('GB', 'IE') (K1)	5	<code>organizations</code> , <code>facilities</code> , <code>suppliers</code> , <code>consultant_profiles</code> , <code>emission_factors</code> (<code><t>country_in_list</code> ; legacy UK/IE variants mapped first)
Currency IN ('GBP', 'EUR') (K2)	8	<code>organizations.currency</code> ,

Group	Count	Target tables / names
		suppliers.payment_currency, document_processing_queue.billing_currency, customer_subscriptions.currency, manual_extraction_batches.currency, consultant_profiles.revenue_currency, consultant_billing.currency, system_settings.default_currency
Non-negative ≥ 0 (K3a/b/d)	27	Emission quantities/factors $\times 3$ (emissions_logs.raw_quantity, emissions_logs.calculated_kg_co2e, emission_factors.co2e_multiplier); supplier per-scope $\times 6$ + supplier_categories.default_emission_factor; usage/money counters $\times 13$ (usage_tracking $\times 5$, customer_subscriptions $\times 4$, consultant_billing $\times 4$); file sizes $\times 4$ (file_attachments.file_size, document_processing_queue.file_size_bytes, processing_queue.file_size_bytes, report_generation_queue.final_report_size_bytes) — $\langle t \rangle_ \langle c \rangle_ \text{nonneg}$
Percentage/confidence 0–100 (K3c)	7	organization_metadata.renewable_energy_percentage, organization_metadata.carbon_offset_percentage, report_generation_queue.progress_percentage, customer_documents.confidence_score, emissions_logs.confidence_score, document_processing_queue.ai_confidence_score, document_processing_queue.ai_mapping_confidence — $\langle t \rangle_ \langle c \rangle_ \text{range}$
Status/role IN-lists (K4)	5	processing_queue_status_in_list, document_processing_queue_status_in_list, customer_documents_status_in_list,

Group	Count	Target tables / names
		organization_members_role_in_list, customer_subscriptions_status_in_list (CHECKs over PG enums by design)
CHECK total	53	
UNIQUEs added (K5, as unique indexes)	7	organization_members_org_user_uniq, consultant_clients_consultant_org_uniq, usage_tracking_org_month_uniq, report_versions_report_version_uniq, suppliers_org_vat_number_uniq (partial), suppliers_org_company_number_uniq (partial), emission_factors_year_activity_country_uniq (natural key: reporting_year, activity_type, country)
UNIQUE dropped (K6)	1	password_reset_tokens(user_id) — closes reset-DoS; token UNIQUE retained
Foreign keys added (F1)	11	emissions_logs_emission_factor_id_fkey, emissions_logs_asset_id_fkey, emissions_logs_unit_fkey (→ units.code), customer_documents_supplier_id_fkey, customer_documents_document_type_id_fkey, customer_documents_org_member_id_fkey, dpq_ai_mapped_facility_id_fkey, dpq_ai_mapped_asset_id_fkey, dpq_ai_mapped_supplier_id_fkey (AI-mapping hints, ON DELETE SET NULL), dpq_emission_factor_used_fkey, messages_conversation_id_fkey — all NOT VALID + VALIDATE; NO ACTION elsewhere; report_versions.report_id intentionally omitted (no reports parent in dump)
NOT NULL tightenings	13	organizations.is_active (C2, in 001); K7 — organization_id on 6 hot tenant tables (conversations, messages, upload_batches, manual_review_queue, file_attachments, customer_verifications); K8 —

Group	Count	Target tables / names
		customer_documents.status, document_processing_queue.statu s/qc_required/customer_approved, processing_queue.queue_statu s/sla_breached (with defaults)

Backfill-then-constrain note: every tightening populates existing rows first (K7 derives org from parent rows where possible; the guarded SET NOT NULL raises and fails safe on any residual NULL). K7's backfill is irreversible — a pre-migration snapshot of the six tables is mandatory (Gate 6 rehearsal).

4. Migration 003 — Indexes

Non-transactional file (CONCURRENTLY): no BEGIN/COMMIT, applied statement-by-statement outside a transaction; each statement independent and idempotent.

Family	Count	Index names
I1 — tenant RLS-path composites	4	customer_documents_org_created_idx, emissions_logs_org_start_date_idx, suppliers_org_idx, facilities_org_idx
I2 — queue-claim partials	3	dpq_claim_idx, processing_queue_claim_idx, report_generation_queue_claim_idx (partial predicates must match worker claim queries exactly)
I3 — messaging/notifications	3	messages_conversation_created_idx, conversation_participants_conv_user_idx, notifications_unread_recipient_idx
I4 — client_access GIN	1	consultant_firm_members_client_access_gin
I5 — trigram (pg_trgm)	3	suppliers_name_trgm_idx, suppliers_vat_number_trgm_idx, organizations_name_trgm_idx
FK-supporting (F1 companions)	4	emissions_logs_emission_factor_id_idx, emissions_logs_asset_id_idx, customer_documents_supplier_id_idx, dpq_customer_document_id_idx
Total	18	(K5's 7 unique-backed indexes were built in 002 and also serve lookups; the rejected I6 blanket programme is absent)

Extensions (2, created IF NOT EXISTS at the top of 003): `pg_trgm` (pre-requisite for I5) and `pgcrypto` (required by `anonymise_user()` in 005 for the sha256 erasure email hash).

5. Migration 004 — RLS

Verify-first and additive only: never drops/alters/weakens an existing policy; ENABLE ROW LEVEL SECURITY where absent; CREATE POLICY only where no policy of the same name exists on the same table. One pattern throughout: `auth.uid()` membership via `organization_members`, evaluated by SECURITY DEFINER helpers (row_security off inside, search_path pinned). No FORCE ROW LEVEL SECURITY. No UK/IE policy differences.

Helper functions (2, created here because policies depend on them): `public.is_org_member(uuid)` — active-membership predicate; `public.is_org_active(uuid)` — tenant-live predicate (C2 suspend switch; write policies require `organizations.is_active = true`). EXECUTE revoked from PUBLIC, granted to `authenticated + service_role`.

Table group	Tables	Policies per table	Naming pattern	Policies
Org-scoped tenant tables	36 (every <code>organization_id</code> -bearing table per the dump, e.g. <code>customer_documents</code> , <code>emissions_logs</code> , <code>suppliers</code> , <code>facilities</code> , <code>organization_members</code> , both queues, messaging, billing, usage)	4 (SELECT / INSERT / UPDATE / DELETE; writes also require <code>is_org_active</code>)	<table>_tenant_select / _insert / _update / _delete	144
organizations (tenant root)	1	2 (member select, member update; no INSERT — org creation is service-role only)	<code>organizations_member_select / _update</code>	2
users	1	2 (self select/update)	<code>users_self_select / _update</code>	2
notifications	1	2 (recipient select/update)	<code>notifications_recipient_select / _update</code>	2
Reference tables (read-only to authenticated; writes service-role)	10 (<code>activity_categories</code> , <code>document_type_categories</code> , <code>document_types</code> , <code>email_templates</code> , <code>emission_factors</code> , <code>glossary</code> , <code>notification_templates</code> , <code>roles</code> ,	1 (SELECT ... USING (true))	<table>_authenticated_read	10

Table group	Tables	Policies per table	Naming pattern	Policies
	supplier_categories, units)			
Total				~160 (create-if-absent; pre-existing policies stand and are validated by the Gate 4 penetration matrix)

RLS is enabled on every table above (36 tenant + organizations + users + notifications + 10 reference); the dangerous “RLS enabled, zero policies” state is surfaced by 007 \$6.

6. Migration 005 — Functions

Function	Purpose	Security posture
<code>public.set_updated_at()</code>	Generic BEFORE UPDATE trigger function stamping <code>updated_at = now()</code> ; companion to 006	SECURITY INVOKER by design (runs with the updating role’s rights)
<code>public.anonymise_user(uuid, uuid, text)</code>	GDPR/UK-GDPR anonymise-in-place erasure — the only approved function-level compliance artefact; launch-gated on the Gate 5 staging rehearsal with residual-PII scan. Derives the marker email from a SHA-256 hash of the user ID — deleted- <code><sha256(user_id)>@anonymised.invalid (005: 'deleted-' encode(sha256(p_user_id::text::bytea), 'hex') '@anonymised.invalid');</code> sets <code>first_name = 'Deleted'</code> and <code>last_name = 'User'</code> (mirroring 005 exactly), nulls <code>password_hash</code> , deactivates the account, scrubs profile PII across <code>consultant_profiles</code> , <code>staff_profiles</code> , <code>beta_users</code> , <code>user_feedback</code> ; preserves the UUID and all FK/audit aggregates (hard delete structurally impossible — ~40 referencing FK columns). Idempotent; transactional; irreversible by design	SECURITY DEFINER, pinned <code>search_path = public</code> ; actor guard (data subject self-service, active staff, or service context); EXECUTE revoked from PUBLIC, granted to <code>service_role + authenticated</code> (guarded self-service)

Explicitly absent: audit hash-chain functions (rejected), retention/pg_cron wrappers and soft-delete functions (deferred).

7. Migration 006 — Triggers

One `trg_set_updated_at_<table>` BEFORE UPDATE trigger per mutable table carrying an `updated_at` column — **up to 76 tables** (reference/config ×11, tenant core ×8, tenant business ×39, staff/QC/operations ×18; candidates missing `updated_at` are skipped with NOTICE, so the actual count is confirmed by 007 \$7). Each trigger is dropped by its deterministic name and recreated (idempotent; touches nothing else).

Deliberately excluded — 6 append-only log tables: `activity_logs`, `document_activity_log`, `email_logs`, `processing_logs`, `user_activity_log`, `review_audit_trail` (the approved v1.0.1 item drops `updated_at` from exactly these; installing maintenance triggers now would bless UPDATES on immutable rows).

8. Migration 007 — Verification

Read-only (SELECTs and NOTICE-only DO blocks); safe against production; run after 001–006, before the Gate 4 penetration matrix and application smoke tests; **re-runnable at any time**. The 10 check areas:

\$	Check area	Pass criterion
1	Renames (R1–R3 + region retirement)	Old names absent; 7 expected new objects present
2	New columns (C1–C10)	16 rows with expected nullability; <code>file_size</code> = bigint; <code>postcode</code> nullable
3	Foreign keys (F1)	11 FKs validated; zero NOT VALID constraints schema-wide; all orphan counts 0; K7 NULL-org counts 0; K6 <code>user_id</code> uniqueness gone, token unique retained
4	Indexes	All 18 from 003 present + 7 K5 unique-backed
5	CHECK constraints (C1, K1–K4)	All validated; country/currency/status violation, negative and out-of-range counts zero
6	RLS (004)	No RLS-enabled-no-policy table; no org-bearing table without RLS
7	Triggers (006) + functions (005)	Trigger coverage complete; append-only exclusions empty; 4 functions present with correct grants
8	Ireland-beta readiness	Eircode path, IE factor insert, GB/IE vocabularies accepted
9	Data preservation	Row counts before/after unchanged
10	One-line summary	NOTICE per gate; any FAIL row blocks sign-off

9. Cross-Migration Dependency Notes

- **Rename-then-constrain ordering:** 002's constraints, 003's indexes, 004's policies and 005's functions all reference the final names produced by 001 (`emission_factors`, `emission_factor_id`, `emission_factor_used`, `default_factor_year`, `eircode`, `is_active`); no later file may run before 001 verifies clean. The RC2-003 rename (`manual_extraction_items.defra_factor_used`) ships inside 001 so all downstream objects see only final names.
- **pgcrypto before first anonymise_user execution:** the extension is created in 003; 005's function creation and any Gate 5 rehearsal invocation must follow 003.
- **K4 vocabulary reconciliation before VALIDATE:** the five status/role IN-lists are scoped from the application enumerations as currently understood; the staging audit must reconcile each list against the application's centralised status constants (adjust the list, not the data) before 002 is validated in production. The same audit gates the K3c 0–100 scale assumption.
- **Seed data is a SEPARATE future package** (Phase B: DEFRA/DESNZ current-year factors + minimal SEAI/EPA Irish core + reference data + seed-user cleanup). It depends on 001–003 only (final names, factor provenance columns, GB/IE vocabularies, factor natural-key uniqueness) — not on 004–006 — and is not part of this manifest's execution batch.
- **I2 partial-index predicates must match worker claim queries exactly** (checked in Gate 7, not enforceable by the migration).

10. Approval Sign-off

Approval of this manifest authorises generation and gated, one-at-a-time execution of migration packages 001–007 exactly as ordered above, subject to Gate 1 inspection as the standing precondition.

☐ Approved — Product Owner, date

Counts verified against migration files 001–007 and release notes 008: 16 columns added; 1 table renamed; 4 column renames + 1 retirement (incl. RC2-003); 1 type widening; 1 NOT NULL relaxation; 53 CHECKs; 7 UNIQUEs added + 1 dropped; 11 FKs; 13 NOT NULL tightenings; 18 indexes (+7 unique-backed from 002); 2 extensions; ~160 RLS policies across 36 tenant tables + organizations + users + notifications + 10 reference tables; 4 functions (2 RLS helpers in 004, 2 in 005); up to 76 triggers with 6 append-only exclusions; 10 verification areas.